

Using the HMRC Test Client

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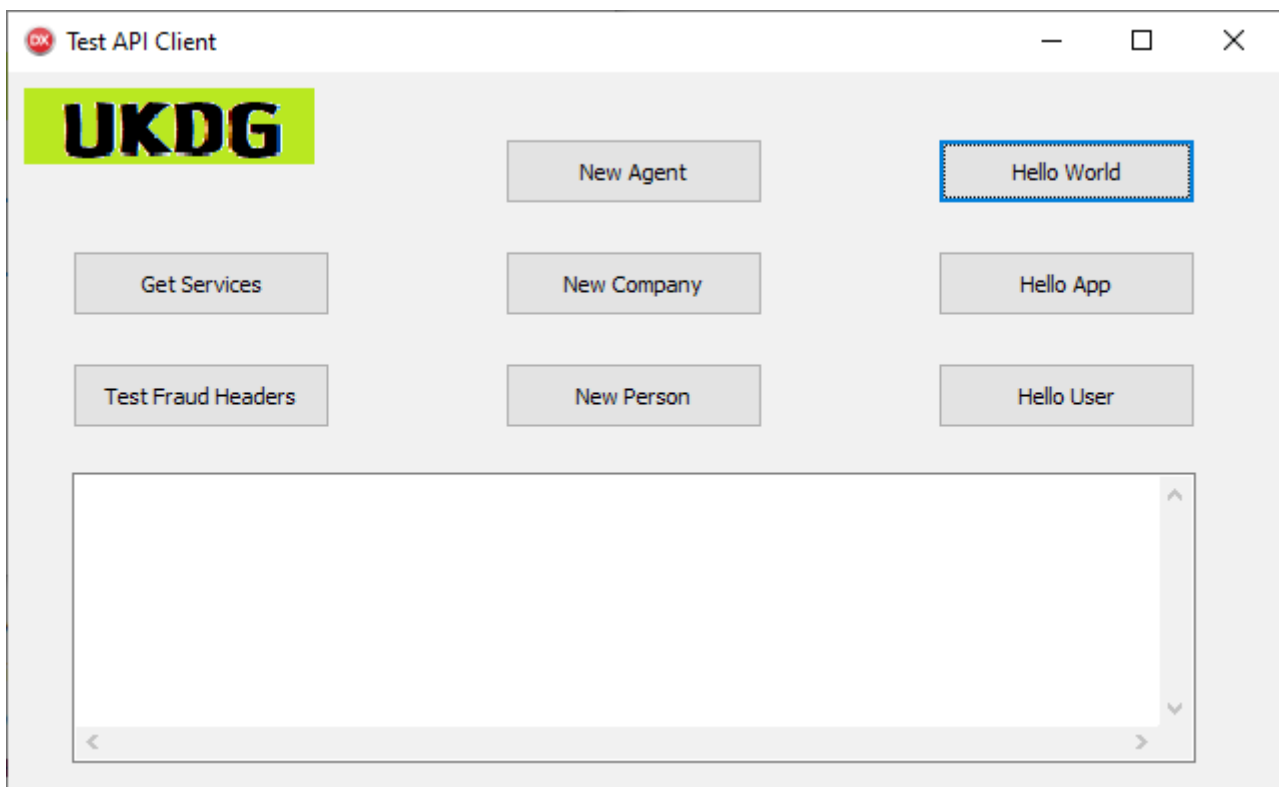
Introduction

This document covers the Test Client app from the UKDG attempt to handle the HMRC MTD APIs. The process started towards the end of 2018, with a coding session at one of the meetings in early 2019 followed by the release of the first version of the API client, then handling only the VAT and test APIs, as they were at that time. The first update was released in December 2020, nearly 2 years after the start, with another update in January 2021, including the updated anti-fraud headers. This document accompanies the third update released in early February 2021, which includes the first release of the Self Assessment (SA) client.

With the development of the SA client, the tests, which had been a part of the VAT demos, have been moved to their own separate module, although note that the VAT demos do still include the original test form. The stand alone test project now includes both them original test demo form and processes, but also the SA setup processes. This might seem strange, but the tests all access endpoints that use application authorisation and that is handled by the Test Client component, while the “real” endpoints all use user level authentication.

It is assumed that the reader is familiar with the HMRC MTD process, so understands the authorisation access levels and the formats of the endpoints which they wish to access. The general or core tests should be relevant to everyone wishing to set up tests in the sandbox. Other tests will be relevant only to those accessing that particular API.

The Core tests.



Accessed from the Core Tests button on the MTDCoreTests application main form. The core tests remain in their original form from the original release, although the services and headers tests were only added in the December 2020 release.

The responses from the endpoints are simply written to the memo and may also be written out to text files.

The 3 Hello buttons on the right of the form access the corresponding Hello endpoints in the test API and basically just show how to authorize and connect, as described in the HMRC tutorials.

The get services button access the services endpoint and lists the available services. The test fraud headers button calls the test headers endpoint. Fraud headers have been a moving target and were updated in December 2020, which is the version targetted by the demos at the time of release.

The 3 “New” buttons create new instances of test users in the HMRC MTD sandbox, as either agents, companies or individuals. The details of what is available can be found in the associated HMRC web pages and what is actually created is in the test client source code. They are created belonging to the client application whose keys were used to created them. See application keys below. At the time of writing, there is still occasionally an issue with calling one of the “new” endpoints after testing the hello user. In this case, just close the form and reopen it. To be resolved.

Typical data returned from the new xyz buttons:

// Person

```
{"userId":"046424627800","password":"xpp0rv57hcwd","userFullName":"Uli Foster","emailAddress":"uli.foster@example.com","individualDetails":{"firstName":"Uli","lastName":"Foster","dateOfBirth":"1946-10-08","address":{"line1":"38 Liverpool Road","line2":"Manchester","postcode":"TS22 1PA"}}, "saUtr":"2904025148", "nino":"KT771329A", "mtdItId":"XAIT00939401724", "eoriNumber":"GB537269878195", "groupIdentifier":"459516456869"}
```

// Company

```
{"userId":"504165559072","password":"QllfRg1q5gqR","userFullName":"Deane Quirke","emailAddress":"deane.quirke@example.com","organisationDetails":{"name":"Company FZESBB","address":{"line1":"23 Warwick Avenue","line2":"Fleet","postcode":"TS25 1PA"}}, "saUtr":"2365446198", "nino":"NL608057B", "mtdItId":"XYIT00529036273", "empRef":"998/BBJI2EWNQS", "vrn":"120220743", "vatRegistrationDate":"2010-02-15", "groupIdentifier":"309200608894"}
```

// Agent

```
{"userId":"466557595571","password":"q1cydrQebhzx","userFullName":"Gale Felix","emailAddress":"gale.felix@example.com", "groupIdentifier":"679445551549"}
```

When using the OAuth2 process to grant authority in the sandbox, the login values required are the **userid** and **password** for the agent/company/person being used for the test case.

For VAT testing, the UID is the **vrn** for the company, for PAYE/Self Assessment it is the **saUtr** or the **nino**. For each testable API in the sandbox, there will be an appropriate ID type and value in the test case.

The SA tests.

UKDG

UTR 1156544395 NINO CH173142D Tax Year 2019-20

Benefits Employments Income Tax

MA Eligibility MA Status Status Transferor National Insurance

```
// Benefits
{"employments":
[{"employerPayeReference": "123\\AB456", "companyCarsAndVansBenefit": 44, "fuelForCompanyCarsAndVansBenefit": 99, "privateMedicalDentalInsurance": 64, "vouchersCreditCardsExcessMileageAllowance": 80, "goodsEtcProvidedByEmployer": 125, "accommodationProvidedByEmployer": 375, "otherBenefits": 16, "expensesPaymentsReceived": 415},
{"employerPayeReference": "456\\AB456", "companyCarsAndVansBenefit": 50, "fuelForCompanyCarsAndVansBenefit": 249, "privateMedicalDentalInsurance": 75, "vouchersCreditCardsExcessMileageAllowance": 23, "goodsEtcProvidedByEmployer": 250, "accommodationProvidedByEmployer": 275, "otherBenefits": 87, "expensesPaymentsReceived": 265}]}
```

At the time of writing there were 7 endpoints for this purpose, although they are not tests in the normal sense, but rather processes to set up test data in the sandbox to be used by the test versions of the SA endpoints.

Most set up standard data for the supplied UTR and tax year, so in general it is just a button clicking exercise. The UTR will be from one of the tests users created with the core tests form. The data created will be displayed in the memo and written to the UserData.txt file. MA Eligibility requires the Nino instead of the UTR.

While the HMRC website lists separate APIs for PAYE, NI and SA, in fact these are all duplicates and all exist in the SA API listing. As a result, the potential separate PAYE and NI client have been dropped for now and everything is handled under the SA heading. The MTD APIs are evolving, so this may change in the future.

Application Keys.

The application keys are generated when the application is created and/or registered with the MTD process and there are separate sets of keys for the sandbox and live environments. The server token no longer exists, but may be referenced in older texts and in the code, to prevent breaking apps that may have used it.

The keys required are the ClientId and ClientSecret. In the various test apps and processes that comprise the UKDG MTD effort, the keys are stored in and loaded from an ini file. This format was used for convenience, being the easiest to manage. In a real business application, they can be stored in any appropriate way. All demos come with a sample ini file, HmrcDevTest.ini, to show the process. This contains a set of expired sandbox keys as an example.

All main forms in all demos load the keys from the selected ini file. The simplest solution is to use the supplied HmrcDevTest file and just replace the key values with the sandbox keys from your registered test application.

Anti-Fraud Headers.

The ini file also contains an AppName and an AppGuid for the fraud headers. Again, these should be updated with values specific to your registered test application. The test app does contain default values, but it may look suspicious if these are used when testing a number of different apps.

The demos only handle the headers for desktop apps and show how these may be set. There are a couple of units that find and/or calculate most of the necessary values, but some must be set manually. See the FormCreate methods of the demo main forms. There are also 2 separate methods for handling them, one in HmrcHeaders and the other in Systematic.VAT.Headers. Developers are encouraged to look at and understand both approaches, although they both accomplish the same thing. The demos use the process in the HmrcHeaders file.

Some Links

The tutorials were here

<https://developer.service.hmrc.gov.uk/api-documentation/docs/tutorials>

The API listing was here

<https://developer.service.hmrc.gov.uk/api-documentation/docs/api/>

The application registration was here

<https://developer.service.hmrc.gov.uk/api-documentation/docs/testing>

The desktop fraud headers were here

<https://developer.service.hmrc.gov.uk/guides/fraud-prevention/connection-method/desktop-app-direct/>